Design of Experiments

Design of experiment is a very broad subject concerned with methods of sampling to reduce the variation in an experiment.

Experimental units

The objects upon which the measurements are taken called experimental units.

Example

Suppose we consider a set of 10 items from the list of supplies to the hospital. Each item is an experimental unit & the set of the items chosen is a sample.

Basic Principles of Design of experiment:

* Randomisation
* Replication
* Local control (error control)

Randomisation:

Randomisation is like insurance. It is always a good idea & even sometimes better than we expect.

Replication

If a treatment is alloted to r-experimental units it is said to replicated r times.

i.e., In a Design of experiment each of the treatment is replicated r times the design is said to be replicated r times.

Local control:

The consideration in regard to the choice of number of replications ensure reduction of standard error of the estimates of the treatment effects because the standard error of the estimate of a treatment effects is $\sigma/\sqrt{r}$. But they cannot reduce the error variance itself, though a large number of replications can ensure a more stable estimate of the error variance. It is however possible to device methods for reducing the error variance. Such measures are called error control (or) local control.

Now we study about some simple & commonly used designs.

There are three such designs.

They are (1) Completely Randomized design (CRD)

(2) Randomized block design (RBD)

(3) Latin square design. (LSD)

Before discussing the above three types, we will discuss about Analysis of variance.

Analysis of variance

Analysis of variance is a technique used to test equality of means, when more than two populations are considered.

In z-test & t-test we considered only the equality of two population means. If there are more than two populations for testing the equality of their means the analysis of variance method is applied

Note:

* R.A. Fisher introduced these ~~techniques~~ techniques.
* This technique originally used in agricultural experiment in which different

types of fertilizers were applied to plots of land, different types of feeding methods to animals and so on.

* Also, it is used to study about average sales by using different sales techniques, the types of drugs manufactured by different companies to cure a particular disease.

* It provides us with a test for the significance of the difference among means.

Assumptions made in the technique:-

1. The samples are independently drawn from the populations.
2. The populations are normally distributed
3. The variances of all the population are equal.

Now, we consider two types of Analysis of Variance:

1. One way classification
2. Two way classification

One-way classification

A set of $N = cr$ observations classified in one direction may be represented as follows.

	X_1	X_2	$\cdots$	X_c
	x_{11}	x_{21}	$\cdots$	x_{c1}
	x_{12}	x_{22}		x_{c2}
	x_{13}	x_{23}		x_{c3}
	$\vdots$	$\vdots$		$\vdots$
	x_{1r}	x_{2r}		x_{cr}
Mean	$\bar{X}_1$	$\bar{X}_2$	$\cdots$	$\bar{X}_c$

Procedure

Step-I : Find N, the total no/- of observations

Step-II : Find T, the total of all observations

Step-III : Find T^2/N, the correction factor.

Step-IV : Calculate the total sum of squares (SST) where

$$SST = \left(\Sigma x_1^2 + \Sigma x_2^2 + \Sigma x_3^2 + \cdots\right) - \frac{T^2}{N}$$

Step-V : calculate the column sum of squares (SSC) where ~~ss~~

$$\cancel{SSC = \frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2}}$$

$$SSC = \left(\frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \frac{(\Sigma x_3)^2}{N_3} + \cdots\right) - \frac{T^2}{N}$$

where N_i denotes the no. of elements in each column for $i = 1, 2 \ldots$

Step-VI :

Prepare the ANOVA Table to calculate F-ratio.

ANOVA Table (Analysis of Variance Table)

Sources of Variation	Sum of Squares	Degrees of freedom	Mean Sum of Squares	Variance Ratio
Between Columns	SSC	$c-1$	$MSC = \frac{SSC}{C-1}$	$F = \frac{MSC}{MSE}$
Within Columns (Error)	SSE	$N-c$	$MSE = \frac{SSE}{N-1}$	(or) $\frac{MSE}{MSC}$
Total	SST	$N-1$		

The F ration should be calculated in such a way that $F > 1$.

Two way Classification

In two way classification of analysis of variance, we consider one classification along column-wise & the other along row-wise.

Procedure For testing two means

H_0: There is no significant difference between column means as well as between row means.

H_1: There is significant difference between column means or between row means.

Step. I : Find N.

Step - II : Find T

Step III : Find T^2/N

Step IV : Find SST (Total Sum of Squares)

Step V : Find SSC (Colum " ")

Step VI : Find SSR (Row Sum of Squares)

Step VII : $SST = SSC + SSR + SSE$

Anova table

Sources of variation	Sum of Squares	Degrees of freedom	Mean Sum of Squares	Variance ratio.
Between Columns	SSC	$c-1$	$MSC = \frac{SSC}{c-1}$	$F_c = \frac{MSC}{MSE}$
Between Rows	SSR	$r-1$	$MSR = \frac{SSR}{r-1}$	$F_R = \frac{MSR}{MSE}$
Residual error	SSE	$N-c-r+1$	$MSE = \frac{SSE}{N-c-r+1}$	
Total	SST	$N-1$		

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Note:

The F-ratio, F_C & F_R should be calculated in such a way that F_C & F_R are greater than 1. Then only it is possible to compare these ratios with the table values.

Problems

1. Set up ANOVA Table for the following per hectare yield for three varieties of wheat each grown in four plots:

Per hectare yield (in hundred kgs)

Plot of land	Variety of wheat		
	A_1	A_2	A_3
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Also work out F-ratio & test whether there is significant difference among the average yields in the 3 varieties of wheat.

Solution

$H_0 : \mu_1 = \mu_2 = \mu_3$ (i.e., The Average yield in the 3 varieties of Wheats are the Same)

$H_1 : \mu_1 \neq \mu_2 \neq \mu_3$

Let us apply F-test to test the hypothesis.

	x_1	x_2	x_3	x_1^2	x_2^2	x_3^2
	6	5	5	36	25	25
	7	5	4	49	25	16
	3	3	3	9	9	9
	8	7	4	64	49	16
Total	24	20	16	158	108	66
	Σx_1	Σx_2	Σx_3	Σx_1^2	Σx_2^2	Σx_3^2

N = No. of observations = 12

T = Total of all observations = $24 + 20 + 16 = 60$

$\frac{T^2}{N}$ = Correction factor = $\frac{60^2}{12} = 300$

SST = Total Sum of squares

$= \left(\Sigma x_1^2 + \Sigma x_2^2 + \Sigma x_2^2 \right) - T^2/N$

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$$= (158 + 108 + 66) - 300 = 32$$

SSC = Colum Sum of Sanares

$$= \left[\frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \frac{(\Sigma x_3)^2}{N_3}\right] - T^2/N$$

$$= \left(\frac{24^2}{4} + \frac{20^2}{4} + \frac{16^2}{4}\right) - 300$$

$$= 144 + 100 + 64 - 300 = 8$$

$$SSE = 32 - 8 = 24$$

Anova table

Source of Variation	Sum of Sanares	Degrees of freedom	Mean Sum of Sanares	Variation ratio
Between Columns	SSC = 8	2	$MSC = \frac{SSC}{2} = 4$	
Error	SSE = 24	9	$MSE = \frac{SSE}{9} = \frac{24}{9}$	$F_c = \frac{4 \times 9}{24} = 1.5$
Total	SST = 32	11		

Test Statistic is $F_c = \frac{MSC}{MSE} = 1.5$

Degrees of freedom = 2, 9

Table value of F at 5% level = 4.26

Conclusion :

Ho is accepted at 5% level since the calculated value of F < the table value of F

$\therefore$ There is no significant difference among the means.

Problem : 2

There are three main brands of a certain powder. A set of 120 sample values is examined and found to be allocated among four groups (A, B; C and D) and three brands (I, II, III), as shown here under.

Brands	A	B	C	D
I	0	4	8	15
II	5	8	13	6
III	18	19	11	13

Is there any significant difference in brands ~~Per~~ Preference? Answer at 5% Level, using one-way ANOVA.

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3) A completely randamised design experiment with 10 plots and 3 treatments gave the following results.

Plot No:	1	2	3	4	5	6	7	8	9	10
Treatment:	A	B	C	A	C	C	A	B	A	B
yield :	5	4	3	7	5	1	3	4	1	7

Two way classification

1) Perform a two-way Anova on the data given below.

		Treatment - I		
		(i)	(ii)	(iii)
Treatment II	(i)	30	26	38
	(ii)	24	29	28
	(iii)	33	24	35
	(iv)	36	31	30
	(v)	27	35	33

Sol use the coding method, substracting 30 from given no/s.

Sol

H_0: There is no significant difference between column means as well as between row means.

H_1: There is significant difference between column means as well as between row means.

The value of F is unaffected by subtracting 30 from all the no/s.

	X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
Y_1	0	-4	8	4	0	16	64
Y_2	-6	-1	-2	-9	36	1	4
Y_3	3	-6	5	2	9	36	25
Y_4	6	1	0	7	36	1	0
Y_5	-3	5	3	5	9	25	9
Total	0	-5	14	9	90	79	102

$N = 5+5+5 = 15$

$T = 0-5+14 = 9$

$C.F = T^2/N = 9^2/15 = 5.4$

$$SST = \left(\Sigma x_1^2 + \Sigma x_2^2 + \Sigma x_3^2\right) - T^2/N$$

$$= (90 + 79 + 102) - 5.4 = 271 - 5.4 = 265.5$$

$$SSC = \left(\frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \frac{(\Sigma x_3)^2}{N_3}\right) - T^2/N$$

$$= \left(0^2/5 + (-5)^2/5 + 14^2/5\right) - 5.4$$

$$= 38.8$$

$$SSR = \left[\frac{(\Sigma y_1)^2}{3} + \frac{(\Sigma y_2)^2}{3} + \frac{(\Sigma y_3)^2}{3} + \frac{(\Sigma y_4)^2}{3} + \frac{(\Sigma y_5)^2}{3}\right]$$

$$- T^2/N$$

$$= \left[4^2/3 + \frac{(-9)^2}{3} + 2^2/3 + 7^2/3 + 5^2/3\right] - 5.4$$

$$= [5.33 + 27 + 1.33 + 16.33 + 8.33] - 5.4$$

$$= 52.92$$

ANOVA Table

Source of Variation	SS	DF	MSS	VR
Between Columns	38.8	2	$\frac{38.8}{2} = 19.4$	$F_C = \frac{21.72}{19.4} = 1.12$
Between rows	52.92	4	$\frac{52.92}{4} = 13.23$	$F_R = \frac{21.72}{13.23} = 1.64$
Error	173.78	8	$\frac{173.78}{8} = 21.72$	
Total	265.5	14		

The Test Statistics are

$$F_C = \frac{MSE}{MSC} = 1.12$$

$$F_R = \frac{MSE}{MSR} = 1.64$$

Degrees of freedom between Columns $= (8, 2)$
Degrees of freedom 〃 rows $= (8, 4)$

Table value of F for $(8,2)$ dof at 5% LOS $= 19.4$
Table value of F for $(8,4)$ dof at 5% LOS $= 6.04$

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Conclusion

In both cases, Calculated value of F
< table value of F

$\therefore$ H_0 is accepted at 5% level. Hence there is no significant difference between the means due to treatment I (or) treatment II.

(2) perform two-way ANOVA for the data given below.

Plots of Land	Treatment			
	A	B	C	D
I	38	40	41	39
II	45	42	49	36
III	40	38	42	42

Use Coding method, Subtracting 40 from the given numbers.

Complete Block Designs

Some simple & Commonly used designs. There are three such designs. They are

1. Completely Randomized design
2. Randomized Block Design
2. Latin Square Design

1. Completely randomized design

Designs are usually characterized by the nature of grouping of experimental units & the procedure of random allocation of treatments to the experimental units.

In a Completely randomized design the units are taken in a single group & in this design no 'local control measure is provided. Also for large no/- of plots (or) treatments this method is not desirable.

~~Note~~ 2. Randomised Block Design

Let us consider an agricultural experiment using which we wish to test the effect of k fertilizing treatments on the yield of a crop. We assume that we know some information about the soil fertility of the plots. Then we divide the plots into 'b' blocks, according to the soil fertility, each block containing 'k' plots. Thus the plots in each block will be homogeneous fertility as far as possible.

Within each block, the 'k' treatments are the given to the 'k' plots in a perfectly random manner, such that each treatment occurs only once in any block. But the same k treatments are repeated from block to block. This design is called Randomised Block Design.

3. Latin Square Design:

We consider an agricultural experiment, in which n^2 plots are taken and arranged in the form of an nxn square, such that the plots in each row will be homogeneous as far as possible with respect to one factor of classification, say, soil fertility and plots in each column will be homogeneous as far as possible with respect to another factor of classification, say seed quality.

Then n treatments are given to these plots such that each treatment occurs only once in each row and only once in each column. The various possible arrangements obtained in this manner are known as Latin squares of order n. This design of experiment is called the Latin square design.

Working Rule

Step-1 Find N

Step: 2 Find T

Step-3: Find $T^2/N = CF$

Step-IV: Find SST

Step V: Find SSC

Step VI: Find SSR

Step VII: Find SSK (Treatment Sum of squares)

Preparation of Anova table

Sources of Variation	Sum of Squares	Degrees of freedom	Mean Sum of Squares	Variation Ratio
Between Columns	SSC	$(K-1)$	$MSC = \frac{SSC}{C-1}$	$F_C = \frac{MSC}{MSE}$
Between Rows	SSR	$(K-1)$	$MSR = \frac{SSR}{K-1}$	$F_R = \frac{MSR}{MSE}$
Between Treatments	SSK	$(K-1)$	$MSK = \frac{SSK}{K-1}$	$F_T = \frac{MSK}{MSE}$
Residual	SSE	$(K-1)(K-2)$	$MSE = \frac{SSE}{(K-1)(K-2)}$	
Total	SST	K^2-1		

The F-ratios F_C, F_R, F_T are calculated in such a way that they are each greater than one.

Note:

For Example, if $\frac{MSC}{MSE} < 1$ then take

$$F_C = \frac{MSE}{MSC}$$

Problems

The following is a Latin Square of design when 4 varieties of seeds are being tested. Set up the analysis of variances table and state your conclusion. You may carry out suitable change of origin and scale.

Sol:

A 105	B 95	C 125	D 115
C 115	D 125	A 105	B 105
D 115	C 95	B 105	A 115
B 95	A 135	D 95	C 115

Solution

Subtract 100 and divide by 5.

	X_1	X_2	X_3	X_4	Total
Y_1	1	-1	5	3	8
Y_2	3	5	1	1	10
Y_3	3	-1	1	3	6
Y_4	-1	7	-1	3	8
Total	6	10	6	10	32

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&

X_1^2	X_2^2	X_3^2	X_4^2
1	1	25	9
9	25	1	1
9	1	1	9
1	49	1	9
20	76	28	28

Arrange the elements in the order of treatment.

					k
A	1	1	3	7	12
B	-1	1	1	-1	0
C	5	3	-1	3	10
D	3	5	3	-1	10

$N = 16$

$T = 12 + 0 + 10 + 10 = 32$

Correction factor $= T^2/N = \frac{32^2}{16} = 64$

$SSR = 8^2/4 + 10^2/4 + 6^2/4 + 8^2/4 - 64$

$= 16 + 25 + 9 + 16 - 64 = 66 - 64 = 2$

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$$SSE = 6^2/4 + 10^2/4 + 6^2/4 + 10^2/4 - 64$$

$$= 9 + 25 + 9 + 25 - 64 = 4$$

$$SSK = \frac{12^2}{4} + 0^2/4 + 10^2/4 + 10^2/4 - 64$$

$$= 36 + 0 + 25 + 25 - 64$$

$$= 22$$

$$SST = 20 + 76\cancel{4}\ 28 + 28 - 64$$

$$= 152 - 64 = 88$$

$$SSE = SST - (SSK + SSC + SSR) = 88 - (22 + 4 + 2)$$
$$= 60$$

ANOVA Table

Source of Variation	Sum of Squares	DOF	Mean Sum of Squares	Variance Ratio
Between Rows	$SSR = 2$	3	$0.67 = MSR$	$F_R = \frac{10}{0.67} = 14.9$
Between Columns	$SSC = 4$	3	$MSC = \frac{SSC}{C-1} = 4/3 = 0.33$	$F_C = \frac{10}{1.33} = 7.52$
Between Treatments	$SSK = 22$	3	$MSK = \frac{SSK}{K-1} = 7.33$	$F_T = \frac{10}{7.33} = 1.36$
Residual Error	$SSE = 60$	$3 \times 2 = 6$	$MSE = 10$	
Total	$SST = 88$	$K^2 - 1 = 15$		

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The table value of F with DoF $(6,3)$

$= 4.76$ (at 5% LOS)

$\therefore$ There is a significant difference between rows as well as between column.

But there is no significant difference between treatments.

Sum - 2

A variable trial was conducted on wheat with 4 variaties in a Latin Sauare Design. The plan of the experiment and the per plot yield are given below:

C 25	B 23	A 20	D 20
A 19	D 19	C 21	B 18
B 19	A 14	D 17	C 20
D 17	C 20	B 21	A 15

Analyse data and interpret the result.